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Project Report

Occlusion-Robust Face Identification Using a Pretrained ArcFace Model

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1. Introduction

Modern face-recognition systems perform well when a frontal face is clear, well illuminated, and similar to the images used during training. Their performance can fall sharply when identity-bearing facial regions are hidden by a medical mask, sunglasses, scarf, hat, or another object, or when lighting and head pose change. These conditions are common in real photographs, but a small laboratory model trained only on clear frontal faces can easily memorise the training images instead of learning stable identity features.

This project addresses the problem as **closed-set face identification**. Every test image is assumed to belong to a person who is already enrolled in the training set. The model receives one face image and returns the corresponding participant ID, such as P001 or P002. The task is different from face-mask detection: a mask detector predicts whether a mask is present, while this project must predict *who the person is* despite the obstruction.

The project uses a consent-based local dataset containing enrolled participants under normal and obstructed conditions – masks, sunglasses, clear glasses, scarves, hats, combined obstructions, and pose and lighting variation. The proposed model is a pre-trained ArcFace face-recognition network with an iResNet50 backbone, paired with a RetinaFace detector for locating and aligning the face, followed by a closed-set identity classifier [1, 2]. This report presents the data collection plan, the consent process, the volume and variation of data collected so far, the cleaning and preprocessing pipeline, and the reasoning behind the chosen models.

2. Problem Statement and Objectives

2.1 Problem Statement

Given an input image containing a normal or partially obstructed face, the system must classify the image as one of the identities present in the training set. Different obstruction types hide different facial regions: a mask removes information from the nose, mouth, and jawline; sunglasses remove much of the eye region; a hat changes the forehead and hairline; and low light reduces visible texture. The central question is whether a model trained on multiple real appearance conditions can learn identity features that remain discriminative when an enrolled person appears with an obstruction, different lighting, or a different head pose. Performance is compared by obstruction type rather than only as one overall figure, because a model can score well on average while still failing on a specific condition such as sunglasses.

2.2 Objectives

1. Collect a structured face-image dataset covering a range of obstruction, pose, and lighting conditions; data collection has been carried out for 25 participants to date.
2. Capture each participant under eight covering conditions: no covering, mask, sunglasses, clear glasses, scarf, hat, mask with sunglasses, and mask with clear glasses.
3. Include five head poses (front, left, right, up, down) together with normal and low lighting.
4. Detect, align, crop, resize, normalise, and quality-check every face image using one reproducible preprocessing pipeline.
5. Train a pretrained ArcFace iResNet50 model, using RetinaFace for detection and alignment, to classify each enrolled participant using the complete collected dataset.
6. Compare the pretrained ArcFace pipeline against a simple convolutional network trained from scratch, to show the benefit of the pretrained approach.
7. Evaluate overall accuracy, per-participant accuracy, and per-condition accuracy in plain, explainable terms.
8. Produce a simple still-image or webcam demonstration that predicts the enrolled participant ID.

3. Related Research Papers

This section contains papers that propose recognition methods, training strategies, evaluation protocols, or core models. Dataset-description papers are separated into the next section.

Research paper	Main contribution	Relevance to this project
ArcFace: Additive Angular Margin Loss for Deep Face Recognition [1]	Introduces an angular-margin loss that learns compact same-person clusters and well-separated different-person clusters in embedding space.	Provides the primary identity representation and classification principle selected for the proposed system. Paper link .
RetinaFace: Single-Shot Multi-Level Face Localisation in the Wild [2]	Detects faces and predicts facial landmarks in one network.	Supports robust face localisation and five-point alignment before identity classification. Paper link .

Research paper	Main contribution	Relevance to this project
When Face Recognition Meets Occlusion: A New Benchmark [3]	Builds WebFace-OCC by adding diverse masks and glasses to identity-labelled faces and shows that occlusion-diverse training improves recognition.	Supports the decision to train with both clear and occluded samples instead of relying only on normal faces. Paper link .
On Recognizing Occluded Faces in the Wild [4]	Evaluates real upper-face occlusion from sunglasses and lower-face occlusion from masks, and reports a stronger real-to-synthetic performance gap.	Justifies testing on real accessories and keeping synthetic occlusion as an optional augmentation rather than the main evidence. Paper link .
Boosting Masked Face Recognition with Multi-Task ArcFace [5]	Extends an ArcFace-based ResNet50 pipeline with masked-face augmentation and an additional mask-related objective.	Shows that an ArcFace-family model can be adapted to masked faces while retaining performance on unmasked faces. Paper link .
Adversarially Learning Occlusions by Back-propagation for Face Recognition [6]	Learns difficult occlusion patterns adversarially and evaluates on masks, sunglasses, scarves, and synthetic occlusions.	Provides evidence that the location and severity of occlusion should be treated as a central robustness factor. Paper link .

4. Related Dataset Papers

The following papers describe datasets rather than new recognition algorithms. The distinction is important because some face-mask datasets contain only mask-condition labels and do not preserve person identity. Such datasets cannot directly train the required multi-class person classifier.

Dataset paper	Dataset content	Use in this project
Human Face Images from Multiple Perspectives with Lighting from Multiple Directions with No Occlusion, Glasses and Hat [7]	For each subject, the dataset contains five lighting levels, seven lighting positions, five subject positions, and three obstruction conditions: none, glasses, and hat. This produces 525 images per subject.	Closest reference for systematic capture of pose, lighting, glasses, and hats. It strongly informs the local collection protocol. Dataset paper link.
Pgu-Face: A Dataset of Partially Covered Facial Images [8]	Contains 896 images from 224 identities, captured with several consumer cameras. Conditions include clean-shaven appearance, sunglasses, facial hair, and facial hair plus sunglasses.	Useful evidence for identity-labelled data, consumer-camera variation, and upper-face obstruction. Dataset paper link.
FaciaVox: A Diverse Multimodal Biometric Dataset of Facial Images and Voice Recordings [9]	Contains 100 participants, 1,800 face images, six masked/unmasked or shield conditions, and three camera zoom levels.	Strong recent reference for consent-based identity-labelled capture under several real lower-face coverings. Dataset paper link.
Face Mask Wearing Image Dataset [10]	Contains 24,916 images divided into correct and incorrect mask wearing, four mask types, and age/gender categories.	Useful for understanding mask diversity. It is not sufficient for person identification because its primary labels are mask-wearing categories rather than stable identities. Dataset paper link.
AR Face Database [11]	Contains more than 4,000 images of 126 identities, with facial expressions, illumination changes, sunglasses, and scarf occlusion.	A standard benchmark for comparing upper-face and lower-face occlusion, and a reference point for the local occlusion categories. Dataset information link.

The public resources are useful for designing the experiment and selecting a pretrained model, but they do not replace the local dataset. The final classifier must recognise the identities enrolled by the project group, and those identities do not exist in the public datasets.

5. Data Collection Plan

Every participant is photographed under the same fixed set of conditions, following the same order, so no participant ends up with easier or harder photos than another.

5.1 Pose, Lighting, and Obstruction Categories

Table 3: Head-pose labels used during capture.

Label	Meaning
front	Looking directly at the camera.
left / right	Head turned to the participant's left / right.
up / down	Chin tilted upward / downward.

Table 4: Lighting labels used during capture.

Label	Meaning
normal	Even indoor lighting; the face is clearly visible.
low	Dim lighting; the face is dimmer but still detectable.

Table 5: Obstruction (covering) labels used during capture.

Type	Labels
Single covering	none, mask, sunglasses, clear_glasses, scarf, hat
Combined covering	mask_sunglasses, mask_clear_glasses

5.2 Planned Number of Photos per Person

Every combination of the five poses and two lighting settings (ten combinations per obstruction type) is photographed a fixed number of times. For the more frequently confused conditions – none, mask, sunglasses, and mask with sunglasses – each combination is captured three times, giving 30 photos per condition. For the remaining conditions – clear glasses, scarf, hat, and mask with clear glasses – each combination is captured twice, giving 20 photos per condition.

Table 6: Planned photos per participant, by covering condition.

Covering condition	Photos per person
None (clear face)	30
Mask	30
Sunglasses	30
Clear glasses	20
Scarf	20
Hat	20
Mask + sunglasses	30
Mask + clear glasses	20
Total per person	200

Photo counts differ between *conditions* because some conditions are given more pose-lighting combinations than others, but the same schedule is followed for every participant, which keeps the dataset balanced across identities.

6. Data Collected So Far

Data collection has been carried out for 25 participants. Because a small number of photos are naturally lost to blur, poor framing, or unusable lighting during capture, the number of accepted images varies from person to person, ranging between 180 and 200 images per participant.

Table 7: Accepted images collected per participant.

Participant	Images	Participant	Images	Participant	Images
P001	187	P010	195	P019	185
P002	194	P011	191	P020	196
P003	182	P012	184	P021	192
P004	199	P013	200	P022	180
P005	190	P014	186	P023	199
P006	183	P015	193	P024	187
P007	197	P016	181	P025	191
P008	188	P017	198		
P009	180	P018	189		
Total accepted images					4,747

Because photo counts are not identical across participants and some conditions are naturally harder to capture cleanly than others (for example, sunglasses over a mask can obscure the face-detection check more often than a clear frontal shot), the accepted images do not split *exactly* in the same proportions as the collection plan in Table 6, though they stay close to it. The observed distribution is:

Table 8: Accepted images collected so far, by covering condition (all 25 participants combined).

Covering condition	Images collected	Share of dataset
None (clear face)	730	15.4%
Mask	690	14.5%
Sunglasses	721	15.2%
Clear glasses	489	10.3%
Scarf	456	9.6%
Hat	482	10.2%
Mask + sunglasses	701	14.8%
Mask + clear glasses	478	10.1%
Total	4,747	100%

7. Consent Form

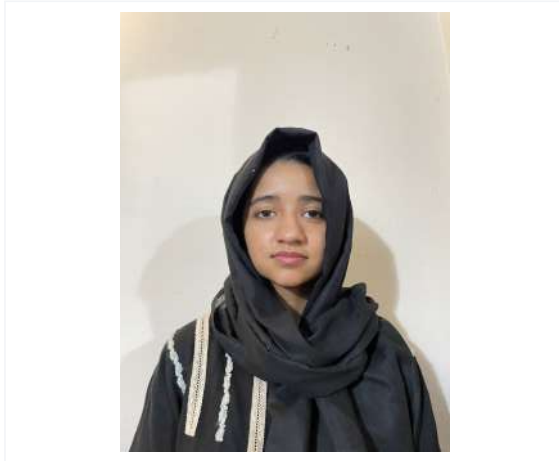
Every participant signs a consent form before any photograph is taken. In brief, the form explains that this is an academic prototype and not a surveillance or security product; that each participant is assigned an anonymous ID (such as P001) so that no name, student ID, or phone number is ever used as a label; that the raw images and identity mapping are kept in access-restricted storage available only to the six project members and are never published or shared outside the group; that participation is voluntary and a participant may withdraw their data at any time before 30/08/2026 with no penalty; and that a recognisable photo of a participant’s face is only shown in the report, presentation, or demonstration if that participant separately ticks an optional permission box. Signing the form confirms the participant is at least 18 years old and has voluntarily agreed to these terms.

8. Sample Images and Observed Data Variation

To make sure the model does not just memorise “easy” photos, the dataset deliberately contains real variation: different coverings, camera angles, and lighting. Figure 1, at the end of this section, shows examples actually captured; their condition labels are given below.

Table 9: Labels for the sample images in Figure 1.

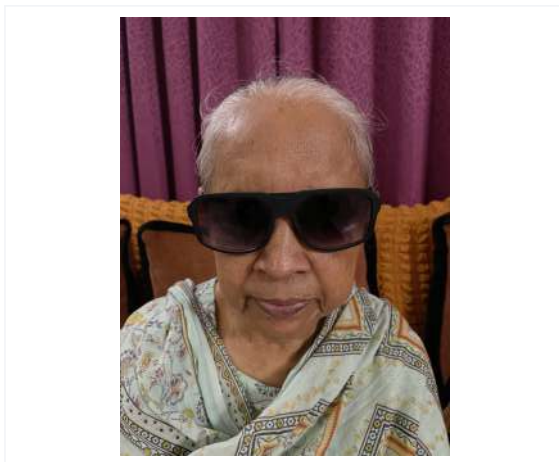
Sample	Covering	Pose	Lighting / notes
(a)	scarf	front	normal light; lower face hidden.
(b)	sunglasses	near-front	normal light; eye region hidden.
(c)	sunglasses	near-front	different device; extra camera variation.
(d)	mask	side turn	normal light; lower face hidden and pose-rotated.
(e)	hat	side turn	normal light; hairline/forehead hidden.
(f)	none	front	low light; borderline-acceptable brightness.



(a) Scarf, frontal pose, normal light.



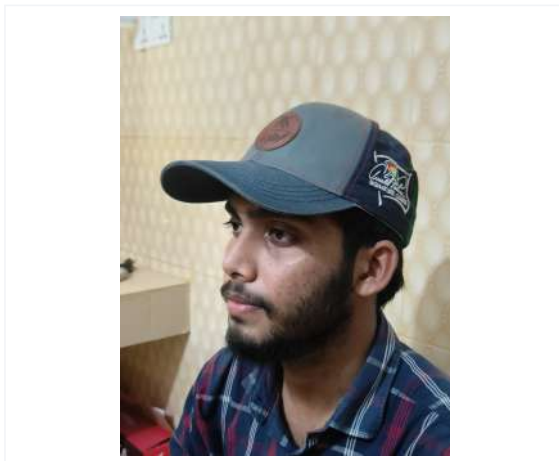
(b) Sunglasses, near-frontal pose.



(c) Sunglasses, second device/angle.



(d) Mask combined with side pose.



(e) Hat combined with side pose.



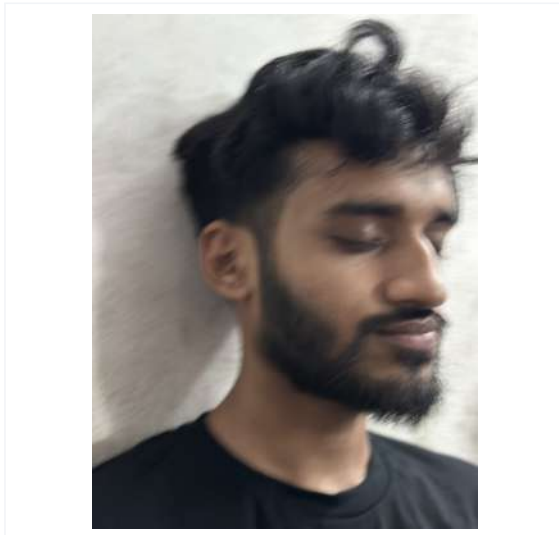
(f) Clear face, low-light condition.

Figure 1: Real samples showing the covering, pose, and lighting variation captured for this project. Only anonymous condition labels are attached to each photo; no participant name is stored with the image.

9. Data Cleaning and Quality Control

Not every captured photo is usable. Before a photo is accepted into the dataset, it is checked against a small set of rules, and anything that fails is set aside with the reason recorded rather than silently deleted:

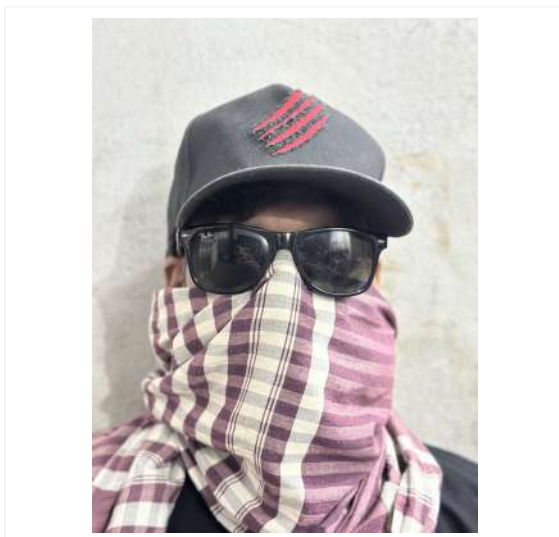
- **Too blurry:** the face is not sharp enough to see clear features.
- **Too dark / too low light:** the lighting is so poor that facial features cannot be made out at all, unlike the merely-dim “low” lighting condition, which is kept.
- **Face fully covered:** the covering hides so much of the face that no identity-bearing region remains visible.
- **Badly framed / mis-cropped:** the face is cut off, too small, or positioned so a usable crop cannot be produced.
- **Duplicate or near-duplicate shots:** accidental repeats of the same pose, lighting, and covering combination.
- **Wrong or missing label:** the recorded pose, lighting, or covering does not match what is actually in the image.



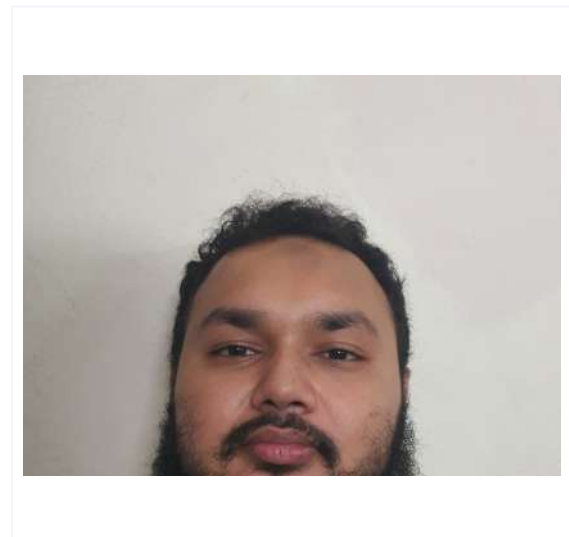
(a) Blurred – motion or focus blur makes facial features indistinct.



(b) Too low light – too dark for facial features to be identified.



(c) Fully covered – no identity-bearing facial region remains visible.



(d) Mis-cropped – the face is not useably positioned in the frame.

Figure 2: Real examples of photos that fail quality control and are excluded from the training and test sets.

Excluded photos are kept in a separate rejected folder together with their reason, so borderline decisions can be reviewed by the team before the dataset is finalised.

10. Data Preprocessing

Every accepted photo goes through the same fixed sequence of steps before it is used for training or testing:

1. **Check the label:** confirm the participant ID, obstruction, pose, and lighting information is present and correct, and that the file opens.

2. **Find the face:** RetinaFace locates the face and marks five key points – both eyes, the nose, and both mouth corners [2].
3. **Straighten the face:** the photo is rotated and scaled slightly so the landmark layout matches a standard reference position, removing differences caused by camera tilt or head roll.
4. **Crop and resize:** a square close-up of the face is produced and resized to 112×112 pixels, the standard input size for ArcFace.
5. **Normalise brightness/colour:** pixel values are adjusted to a standard range so photos from different phones and lighting conditions are more directly comparable.
6. **Final quality check:** any photo where the face cannot be found or aligned is flagged and excluded, using the same documented-rejection approach described in Section 9.
7. **Training-only variation (augmentation):** for training photos only, mildly modified copies are created – for example a mirrored, slightly brighter/darker, or slightly rotated version – to give the model more variety to learn from. These modified copies are never used for validation or testing.

10.1 Data Organisation and Split

A structured filename such as

P003_mask_left_normal_01.jpg

is used for quick inspection, while a CSV file remains the authoritative metadata source, containing at least:

image_path, person_id, occlusion, pose, lighting, device_id,
quality_flag

Each photo is a separate, deliberately composed capture rather than a burst-mode frame, so images are split into training, validation, and test sets using a fixed random 70/15/15 split, applied separately for each participant and each covering condition. This keeps every participant and every condition represented in all three sets while still giving the model unseen photos to be validated and tested on.

11. Model Selection

11.1 Selected Models

The recognition pipeline uses two purpose-built models in sequence, plus one simple baseline for comparison:

- **RetinaFace** – a face-detection and landmark model used to locate the face and its five key points, and to straighten it before recognition [2].
- **Pretrained ArcFace (iResNet50 backbone)** – the main identification model. It turns each aligned face crop into a 512-number identity “fingerprint”. Two photos of the same person produce very similar fingerprints, while different people produce clearly different ones; a new photo is identified by finding the enrolled participant whose fingerprint is the closest match [1].
- **Small CNN trained from scratch** – a simple baseline model, trained only on the local dataset, used purely to demonstrate the advantage of the pretrained ArcFace pipeline over building a face model from nothing.

11.2 Why These Models Were Selected

- **Purpose-built for faces:** RetinaFace and ArcFace are designed specifically for finding and telling apart human faces, unlike a general-purpose image classifier [1, 2].
- **Works with limited local data:** because ArcFace already understands faces broadly, it needs far less new data to learn our specific participants than training from scratch.
- **Practical to run:** the iResNet50 backbone is light enough for laboratory hardware while remaining much stronger than a small CNN trained from random initialisation.
- **Proven on occluded faces:** prior published work has already adapted this same model family to masked and occluded faces, supporting its use here [3, 5].

The small from-scratch CNN is kept only as a baseline: with a limited number of identities, it risks memorising backgrounds, clothing, or camera artefacts instead of learning the face itself, which is exactly what the pretrained approach is chosen to avoid.

11.3 Training Approach

The model is trained once, using the complete collected dataset (all participants and all covering, pose, and lighting conditions together) to classify each enrolled participant. Training proceeds in two stages: first the pretrained ArcFace backbone is kept frozen while a new identity layer is trained for the local participants; then the final backbone layer is unfrozen and trained further with a much smaller learning rate, allowing limited adaptation to the local masks, scarves, sunglasses, hats, lighting, and camera characteristics.

12. Evaluation and Testing Plan

The model’s predictions on the held-out test photos are checked against each photo’s true, known identity, using measures that stay in plain language:

- **Overall correctness:** the percentage of test photos identified correctly.
- **Fairness across participants:** whether accuracy is roughly even across participants, or noticeably worse for some.
- **Who gets mixed up with whom:** a simple table showing which participants are most often confused with each other.
- **Accuracy by condition:** accuracy calculated separately for each covering type, and separately for normal vs. low light and for each head pose, to show exactly which conditions are hardest.
- **Face-detection reliability:** how often the face-finding step fails, since a face the system cannot locate can never be identified.
- **Comparison with the baseline:** how much better the pretrained ArcFace pipeline performs than the simple from-scratch CNN.

Testing proceeds by training the model on the full dataset, evaluating it on the held-out test split with the measures above, and manually reviewing a sample of incorrect predictions to understand likely causes – such as heavy blur, an extreme head angle, poor light, or two participants who look similar even to a human. All preprocessing and training choices are finalised before the test split is evaluated; afterwards, no test photo is changed, removed, or replaced except to correct a genuine, documented labelling mistake applied consistently.

13. Conclusion

This report sets out a closed-set identity-classification task under real facial obstruction, a fixed per-participant capture schedule covering eight obstruction conditions across five poses and two lighting settings, and a consent process that protects participant identity while allowing optional, explicit permission for showing example photos. Data collection has reached 25 participants with roughly 180 to 200 accepted images each, for a combined dataset of about 4,747 images after cleaning. The chosen pipeline – RetinaFace for detection and alignment, and a pretrained ArcFace iResNet50 model for identification – is well suited to this scale of data because it starts from strong general face knowledge rather than learning faces from nothing, and it is compared against a simple from-scratch CNN baseline to demonstrate that advantage. The remaining work is to complete preprocessing of the full dataset, train the model on the complete

set of collected images, and evaluate it using the plain-language measures described in Section 12.

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